

Supplementary: Framework API Reference

Overview

This supplementary material documents the core framework modules that implement the theoretical constructs from Part 1. The complete source code is available at:
https://github.com/docxology/cognitive_integrity

Trust Module

The trust module implements bounded trust delegation with configurable decay.

Table: Trust module API: Core classes for trust computation and management. {#tab:trust-api}

Class	Description
TrustCalculus	Computes composite trust: $T = \alpha \cdot T_{base} + \beta \cdot T_{rep} + \gamma \cdot T_{ctx}$. Implements delegation decay: $T_{delegated} = \min(T_{i \rightarrow j}, T_{j \rightarrow k}) \cdot \delta^d$
TrustMatrix	Manages pairwise trust between n agents with $O(1)$ lookups and $O(1)$ updates. Supports efficient path trust queries.
ReputationTracker	Tracks time-decayed reputation based on interaction history.

Firewall Module

The firewall module implements multi-stage classification for cognitive attack detection.

Table: Firewall module API: Classes for message classification and threat detection. {#tab:firewall-api}

Class	Description
CognitiveFirewall	Three-tier classifier (ACCEPT/QUARANTINE/REJECT) with configurable thresholds. Combines pattern matching, semantic analysis, and anomaly detection.
PatternDetector	Heuristic pattern matching with 15 injection patterns and 20 suspicious indicators. Weighted scoring based on pattern severity.
SemanticSimilarityDetector	Embedding-based similarity to

Consensus Module

The consensus module implements Byzantine-tolerant agreement protocols.

Table: Consensus module API: Classes for Byzantine-tolerant multiagent decisions. {#tab:consensus-api}

Class	Description
ByzantineConsensus	Core consensus with $n \geq 3f + 1$ guarantee. Implements three-phase protocol: collect, echo, decide.
WeightedByzantineConsensus	Trust-weighted voting where high-trust agents have greater influence. Prevents low-trust Sybil attacks.
ConfidenceByzantineConsensus	Votes weighted by agent confidence in their own belief.
CombinedByzantineConsensus	Multiplies trust and confidence weights for robust aggregation.

Detection Module

The detection module implements statistical anomaly and drift detection.

Table: Detection module API: Classes for belief drift and anomaly detection. {#tab:detection-api}

Class	Description
DriftDetector	KL-divergence based belief distribution drift detection. Sliding window comparison with configurable thresholds.
AnomalyScorer	Isolation forest anomaly scoring for belief state vectors. Trained on baseline distribution.

Provenance Module

The provenance module implements information flow tracking with causal attribution.

Table: Provenance module API: Classes for belief origin tracking and taint propagation. {#tab:provenance-api}

Class	Description
ProvenanceChain	Linked list of provenance records tracking belief transformations.
ProvenanceGraph	DAG structure for complex multi-source belief provenance. Supports transitive queries.
TaintLabel	Labels for marking untrusted information sources. Propagates through belief operations.
CausalAttribution	Attributes beliefs to original evidence with contribution weights.

Sandbox Module

The sandbox module implements belief partitioning for provisional information management.

Table: Sandbox module API: Classes for belief sandboxing and promotion. {#tab:sandbox-api}

Class	Description
SandboxManager	Manages verified and provisional belief partitions. Enforces TTL expiry and consistency checks.
BeliefPartition	Container for beliefs with shared trust properties. Supports batch operations.
PromotionCriteria	Configurable criteria for promoting beliefs from provisional to verified.

Tripwire Module

The tripwire module implements canary belief monitoring for intrusion detection.

Table: Tripwire module API: Classes for canary belief monitoring. {#tab:tripwire-api}

Class	Description
CognitiveTripwire	Monitors canary beliefs for unauthorized modifications. Configurable alert severity levels.
Canary	Individual canary belief with expected value and tolerance.
TripwireAlert	Alert record with severity, timestamp, and drift magnitude.

Invariants Module

The invariants module implements runtime behavioral constraint checking.

Table: Invariants module API: Classes for behavioral invariant enforcement. {#tab:invariants-api}

Class	Description
InvariantChecker	Evaluates agent actions against registered invariants. Returns violations with severity.
RuntimeMonitor	Continuous monitoring of agent behavior for invariant violations. Supports real-time alerting.
Invariant	Declarative invariant specification with predicate and severity.